

Chapter 2

Types of connectives

1) $P \wedge Q$

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

2) $P \vee Q$

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

3) $P \rightarrow Q = \sim P \vee Q$

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

4) $P \leftrightarrow Q = (P \rightarrow Q) \wedge (Q \rightarrow P)$

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Same value $\rightarrow$ True

5) Tautology (True only)

P	$P \vee \sim P$
T	T
F	T

6) Contradiction (False only)

P	$P \wedge \sim P$
T	F
F	F

7) Contingency (either True or False)

P	Q	$\sim(P \wedge \sim Q)$
T	T	T
T	F	F
F	T	T
F	F	T

Converse, Inverse, Contrapositive, Negation

Given $P \rightarrow Q$

Converse: $Q \rightarrow P$

Inverse: $\sim P \rightarrow \sim Q$

Contrapositive: $\sim Q \rightarrow \sim P$

Negation: $P \wedge \sim Q$